

INTERIM REPORT

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Introduction

Diabetes mellitus is a global chronic condition and an essential factor of mortality that affects humans of all age groups. A key statistic by the international diabetes federation (IDF) stated that around 415 million individuals were living with diabetes globally in 2015, with a prediction of this number rising to 640 million by 2040 (International Diabetes Federation 2015). In the US specifically, diabetes affects around 30 million Americans, with costs rising to \$327 billion in 2017, with \$237 billion going for direct medical costs and \$90 billion going for reduced productivity (American Diabetes Association 2018).

Diabetes has two primary types, Type1, which occurs when the immune system attacks the hormone and the cell that produces the insulin in the pancreas needed to regulate blood sugar levels 2040 (International Diabetes Federation 2015) and Type 2, which occurs due to insulin resistance, which is usually traced back to poor diet or lifestyle, genetics and hereditary factors. (Type 2 diabetes - Symptoms and causes n.d.)

As the number of diabetes patients increases, the number of hospital inpatient readmissions increases accordingly, which is when a patient is readmitted to the hospital after being discharged, inpatient readmissions is usually based the length of readmission timing at the hospital when readmitted, which makes the length of readmission timing also a crucial element in addition to inpatient readmission for managing and optimising resource allocation for best healthcare quality for diabetic patients (Hu and Sokolova 2020). Accidental readmission increase both patient financial strain as well as inefficient use and misallocation of medical resources that is why and in our current era due to the rapid growth that the AI field has experienced, including machine learning and deep learning, healthcare field was one of the primary beneficiary fields from this growth, with these new technologies it has become easier to predict inpatient hospital readmission based on Electronic Health Record (EHR) which accurately predicts the need of each patient identifying patients who are most in need of the services offered to readmitted patients as well as the length of the readmission timing to differentiate between early and late readmissions, using machine learning and deep learning to predict both inpatient readmission and the length of readmission timing can effectively reduce the poor and overuse of medical resources, reduce financial costs of the healthcare system as well as plan according to the prediction for optimal care delivery (Sosa and Hernandez 2023).

Thus, in our study, we will be focusing on predicting inpatient readmission along with the length of readmission timing using machine learning techniques. Our study will also include clustering analysis to discover whether distinct patient subgroups can be identified based on shared characteristics, as well as a data reduction analysis to uncover the most informative features of our dataset to simplify dimensionality while keeping key features.

This research will be guided by 3 principal research questions:

- RQ1: Can data reduction techniques effectively reduce the complexity of our dataset while preserving the most relevant information?
- RQ2: Can unsupervised clustering identify subgroups of diabetic patients with shared characteristics?
- RQ3: How effectively can a two-stage hurdle modelling approach predict hospital readmission and distinguish early (<30 days) versus late (>30 days) readmissions among diabetic patients?

There are 3 core problems that need to be addressed in this study:

- High dimension and mixed types of data: the dataset contains 50 features mixing numerical and categorical variables, as well as features with low variance, missing values and outliers. As there are more categorical variables in our dataset, we will need to pay attention during the feature engineering part.
- Clustering and data reduction related issue: as mentioned in the first problem, our dataset has a mixture of numerical and categorical data, thus there is a crucial need to explore new clustering and data reduction techniques that handle both mixed data types at the same time or techniques that handle the numerical and the categorical subsets of the data.
- Prediction related issue: the target variable of my dataset is “readmitted” which has 3 outcomes, “NO”, “< 30” (early) and “>30” (late), single stage models would treat these 3 outcomes as independent options, and will ignore the fact that a patient cannot have a readmission time if they are never readmitted, this approach biases the models towards “NO” outcome as it is the majority class which makes the model ignore the “<30” class that is why we will be trying the dependent two stage hurdle model to see if it mitigates this issue which models readmission first and then models readmission timing conditional on readmission being “yes”.

This project is relevant to the data analytics field since it will be applying the following:

- Simplifying the data using data reduction techniques to keep only key variables to build a stronger model for clustering, as clustering works poorly on high-dimensional data.
- Identifying Clusters of diabetic patients for better targeted interventions.
- Demonstrating predictive modelling of machine learning and potentially deep learning techniques in healthcare.
- Providing crucial insights for the healthcare industry for better resource allocation, management and decision-making and planning when it comes to readmissions.

The core technologies and research processes that will be used for this study are:

- Data preparation: including the cleaning of the data, dealing with outliers, missing values and duplicates, removing zero variance and low variance features.
- EDA: which will include exploring the dataset and correlations between variables.
- Data reduction analysis: will be using utilising techniques such as FAMD and MCA, as well as numerical techniques only on the numerical variables.
- Clustering: will be performing clustering on both the original and the reduced data to see which one performs better. We will be using techniques such as k-Means, k-Prototype and LCA.
- Feature Engineering: will be encoding our categorical variables into numerical variables from the original data, as it is better to use the original complete data for prediction rather than the reduced data, to ensure the data is ready for the modelling part
- Predictive modelling: will be using the two-stage hurdle model, using many different machine learning models such as Logistic Regression, Decision Trees, Random Forests, k-Nearest Neighbours, XGBoost, and LightGBM.
- Model evaluation: will evaluate our model based on many metrics, such as accuracy, precision, recall and F1 score.

My hypothesis for a solution is:

- The dataset will be reduced to only key variables without losing meaningful information, which will aid in a higher quality of clustering.
- Clustering will reveal useful subgroups for diabetic patients that will help during the predictive modelling phase.
- Two stage hurdle model will perform well with moderate to high performances.

The rest of my report will include a literature review covering studies that have already used this dataset, an exploration of data and methods where I will detail how I prepared my data, cleaned it and visualised it, and a section for future analysis, where I will detail a plan based on the future work I plan to perform.

Literature Review

Various studies have worked on the *Diabetes 130-US Hospitals for Years 1999–2008* dataset. In this section, we will be covering 10 literature reviews.

(Hu and Sokolova 2020) assessed 6 machine learning models including Random Forest, decision trees, SVM and Logistic Regression for a multi-class prediction of readmission which is a three category variable, no readmission, readmission within 30 days, and readmission after 30 days, the authors evaluate the performance of each of the machine learning models based on accuracy, macro recall precision and F1 score, the use of

macro recall was particularly effective to evaluate the models due to the imbalance of the classes. Gradient Boosting and Random Forest achieved the highest accuracies with 64.769% and 64.41% after parameter tuning of 10-fold cross-validation. Despite the good accuracy, one of the limitations or challenges identified in this study is the difficulty of distinguishing between patients with no readmission and readmission within 30 days, since patients of these two classes usually share many similar characteristics. The authors suggest that deep learning methods can mitigate this issue.

Almost the same approach was repeated by (Sosa and Hernandez 2023) but it focuses on a wider range of approaches with other machine learning models including K-Nearest neighbour Naïve Bayes Artificial Neural Networks Multilayer Perceptron (MLP), XGBoost, Genetic Algorithms, fuzzy rule-based systems, and XGBoost this study concludes that Random Forest achieved the highest metrics including accuracy of 68% and precision and recall followed by decision trees 64% accuracy one of the limitations of this study is the modest accuracy levels ranging from 33% to 68% which shows that model is facing challenges to predict.

A similar paper by Sah Kanu and Khanal (2023) uses many machine learning models, including Logistic Regression, decision trees, and Random Forest, to predict readmissions. This study uses Hadoop and PySpark for data preprocessing. Random forest showed the highest scores with 1 on all of the evaluation metrics, including accuracy and precision, followed by decision trees and then logistic regression. The study found that if patients are readmitted within 30 days, it means that the treatment wasn't effective or proper, and the patients with the highest likelihood to be readmitted are patients prescribed diabetes medicines. One of the critical limitations of this study is the high risk of overfitting raised by the perfect scores of the evaluation metrics, which needs more rigorous investigation.

(Shang et al. 2021) evaluated a similar approach it used Smote to fix the class imbalance on the data, and it used 3 models to predict readmissions, Random forest, Naïve bayes and tree ensemble, the models were evaluated based on AUC, random forest achieved the highest AUC, with 0.6, some of the limitations of this study is the modest AUC values of models, in addition to its relying on traditional machine learning models which might struggle to capture complex data.

The paper by (Mahmoud, Bader and McNicholas 2023) divided the study into two scenarios, the first one that predicts all readmission within 30 days or after 30 days, and the second scenario that predicts the readmission cases within 30 days only, which was the only study that correctly identified that the two distinct patient stratification, the dataset was pre-processed using Smote for class imbalance some of the models used were Logistic Regression, decision trees, Random Forest, KNN, Naïve Bayes, SVM, and Adaboost the evaluation metrics used to evaluate this paper are AUROC and accuracy, the first scenario shows that SVM has the highest accuracy of 64.5% and the second

scenario shows that Random Forest has the highest accuracy of 86.3%. One of the limitations of this study is the moderate AUROC values, which show that models are struggling to predict correctly, as well as the small size of the dataset after class balancing, which might have negatively affected the model learning.

(Zarghani 2024) uses the same dataset but compares between traditional machine learning models and Long Short-Term Memory (LSTM) neural network which a deep learning technique, the author predicts a binary classification for the readmissions, many traditional machine learning models were used such as LightGBM, XGBoost, Cat Boost, Random Forest, and Decision Tree, where LightGBM achieves the highest accuracy of the traditional models, but LSTM achieves the global highest accuracy of 97.65%, however the LSTM model is shown to suffer from overfitting, uncovering the trade-off of deep learning techniques, despite the high accuracies, the model might not be as good as the accuracy states.

(Liu et al. 2024) evaluated the same approach as (Zarghani 2024), in which the authors compared deep learning models and traditional machine learning models, in this study the authors again use a binary classification for the readmissions predictions, this study evaluated 10 traditional machine learning models Random Forest (RF), XGBoost, Support Vector Machines (linear and RBF), Decision Tree, Logistic Regression, KNN, Naïve Bayes, AdaBoost, and MLP with the deep learning model Long Short-Term Memory (LSTM), Random forest and XGBoost were the models with the highest accuracy of 0.88, however the LSTM model achieved a lower accuracy of 0.77 despite its complexity, this results contradict the study done by (Zarghani 2024) and the universal claim of deep learning models always outperform machine learning models.

(Santos de Freitas 2023) evaluate the same approach (Liu et al. 2024) and (Zarghani 2024), comparing machine learning models and deep learning model but it predicts the hospital length of stay using Random Forest, Support Vector Regression (SVR), K-Nearest Neighbours Regressor (KNNr) and deep learning techniques including artificial Neural Networks (ANN) and TabNet, where TabNet achieved global highest R-squared value of 0.69 followed by SVR with 0.67 R-squared value, which agrees with (Zarghani 2024) paper and contradicts (Liu et al. 2024) paper in terms of the comparison between the machine learning and deep learning models, however this study showed some limitations such as the increase of prediction error with long hospital stays as well as that TabNet is computationally expensive and needs large labelled datasets to perform well.

(Pham et al. 2019) evaluated a different approach using predictive modelling along with cluster analysis to predict readmissions as a binary variable, the authors applied clustering on patients before feeding them into the classifiers, using K-Means technique on 3 components derived using PCA which identified 4 patient groups, the classifiers included Decision Trees (CHAID, CART), Neural Networks (with bagging), and Tree-Augmented Naïve Bayes networks, where ensemble model achieved the highest

accuracy of 63.5%. This is one of the very few studies that used cluster analysis and data reduction analysis; however, the prediction of the variables is binary, as in the rest of the papers.

(Kaliappan et al. 2024) investigated diabetes predictions across multiple datasets, comparing traditional machine learning models with ensemble and hybrid approaches. The study showed that using feature selection with ensemble learning improves predictions across different datasets most of the time; however, in some other datasets, it did not enhance the performance, because it led to the loss of important information.

While all of these studies present valuable information, none of them have used a two-stage (hurdle) model, along with using cluster analysis and data reduction analysis, all papers focus on a binary classification, and uses cluster analysis and reduction analysis very rarely and very briefly, accordingly we decided to accord more importance to these overlooked techniques that have a high potential of enhancing predictive performance.

Exploration of Data and Methods

The dataset used in this study is the **Diabetes 130-US Hospitals for the years 1999-2008**. It represents 10 years of data with 50 attributes or columns and 101,766 instances or rows, this dataset is publicly available in the **UC Irvine Machine Learning Repository**, and was originally curated and donated by Strack et al. (2014), it contains detailed information about patients with a focus on diabetes from 130 hospitals in the united states including 18 hospitals from Midwest, 58 from Northeast, 28 from South, and 16 from West, this data was collected from electronic health records (EHR) which has demographic information and laboratory test results of patients that attended the different hospitals, the target variable is **readmitted** which indicates if the patient was readmitted within 30 days, readmitted after 30 days or not readmitted at all

The dataset was collected using csv format from the **UC Irvine Machine Learning Repository**, and it was cleaned and processed using python and it was explored and analysed using R.

The ethical approval for this research was obtained in accordance with the regulations of the **Dundalk Institute of Technology (DkIT) School of Informatics & Creative Arts**. As this study is an **undergraduate research project**, the completed ethical approval form was submitted to the **Departmental Research Ethics Committee (DREC)** within the School of Informatics & Creative Arts. Find the full ethics form in Appendix A at the end of the paper.

The different variables or attributes that exist in this dataset are:

Column Name	Type	Description
encounter_id	Nominal	Unique ID for each hospital visit
patient_nbr	Nominal	Unique ID for each patient
race	Nominal	Patient ethnicity (Caucasian, Asian, African American, Hispanic, Other)
gender	Nominal	Patient gender (Male, Female, Unknown/Invalid)
age	Nominal	Age grouped in 10-year ranges (0–10, 10–20, ..., 90–100)
weight	Numeric	Patient weight in pounds
admission_type_id	Nominal	Type of hospital admission (e.g., emergency, urgent, elective, newborn, not available)
discharge_disposition_id	Nominal	Outcome at discharge (e.g., discharged home, expired, not available)
admission_source_id	Nominal	Where the patient was admitted from (e.g., physician referral, emergency room, hospital transfer)
time_in_hospital	Numeric	Number of days the patient stayed in the hospital
payer_code	Nominal	Type of insurance or payer (e.g., BlueCross, Medicare, Self-pay)
medical_specialty	Nominal	Speciality of the admitting doctor (e.g., cardiology, internal medicine, surgeon)
num_lab_procedures	Numeric	Number of lab tests performed during the visit
num_procedures	Numeric	Number of medical procedures (excluding labs)
num_medications	Numeric	Count of different medications prescribed during the visit

Column Name	Type	Description
number_outpatient	Numeric	Outpatient visits in the year before this encounter
number_emergency	Numeric	Emergency visits in the year before this encounter
number_inpatient	Numeric	Inpatient visits in the year before this encounter
diag_1	Nominal	Primary diagnosis (first three digits of ICD-9 code)
diag_2	Nominal	Secondary diagnosis (first three digits of ICD-9 code)
diag_3	Nominal	Additional secondary diagnosis (first three digits of ICD-9 code)
number_diagnoses	Numeric	Total diagnoses recorded for the encounter
max_glu_serum	Nominal	Result of glucose serum test (>200, >300, normal, none)
A1Cresult	Nominal	Result of A1C test (>8, >7, normal, none)
metformin	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
repaglinide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
nateglinide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
chlorpropamide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
glimepiride	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
acetohexamide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)

Column Name	Type	Description
glipizide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
glyburide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
tolbutamide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
pioglitazone	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
rosiglitazone	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
acarbose	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
miglitol	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
troglitazone	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
tolazamide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
examide	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
citoglipton	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
insulin	Nominal	Diabetes medication usage or dosage change (up, down, steady, no)
glyburide-metformin	Nominal	Combination drug usage or dosage change (up, down, steady, no)
glipizide-metformin	Nominal	Combination drug usage or dosage change (up, down, steady, no)

Column Name	Type	Description
glimepiride-pioglitazone	Nominal	Combination drug usage or dosage change (up, down, steady, no)
metformin-rosiglitazone	Nominal	Combination drug usage or dosage change (up, down, steady, no)
metformin-pioglitazone	Nominal	Combination drug usage or dosage change (up, down, steady, no)
change	Nominal	Indicates if diabetes medications were changed (change, no change)
diabetesMed	Nominal	Whether the patient was prescribed any diabetes medication (yes, no)
readmitted	Nominal	Readmission status (<30 for readmitted within 30 days, >30 for after 30 days, No for no readmission)

Our cleaning process included multiple steps:

- Handling missing values: I decided to drop multiple columns due to their high percentage of missing values, such as weight with 97% of missing values, medical speciality with 49% of missing values, and finally payer_code with 40% of missing values, as well as because it has low importance in our prediction. Other columns, such as race, diag_1, diag_2 and diag_3, had some missing values as well, with very low percentages between 0.02% and 2.23%, thus I decided to drop the rows with the missing values since that will not affect my data, as it is less than 4% of the whole data.
- Handling outliers: I ensured to check outliers using boxplots for my numerical columns, all outliers looked normal and reasonable, and there was no significant errors within them, some numerical values were highly skewed but they were kept to present the whole dataset with extreme cases included, and that would distort the natural distribution of the data and could lead to loss of important information.
- Handling inconsistencies: I ensured to fix all categorical inconsistencies or mistakes, such as the names of some categories or variables, for example, diag_1, diag_2 and diag_3, where I mapped 700 unique codes to nine broader health categories to reduce the data complexity.
- Handling duplicates: I removed duplicate patient entries since standard machine learning models like Logistic Regression assume that all data points are

independent. If I include the same patient multiple times, their specific health history biases the model. By keeping only the first occurrence, I establish a clean baseline for each patient and prevent data leakage between training and testing sets.

- Handling irrelevant data or zero-variance data: I dropped some columns that had no relevance to my research objectives, such as `encounter_id`, and that are not meaningful to my prediction. I dropped 4 other columns, which had a zero variance, which are `examide`, `citoglipton`, `metformin-rosiglitazone`, `glimepiride-pioglitazone`, as they will not add any information to my predictions.
- Handling incorrect data types: ensured to convert all the remaining columns into their correct data type to ensure their readiness for exploratory data analysis.

The section below will include some of the most interesting visualisations and analyses of my dataset.

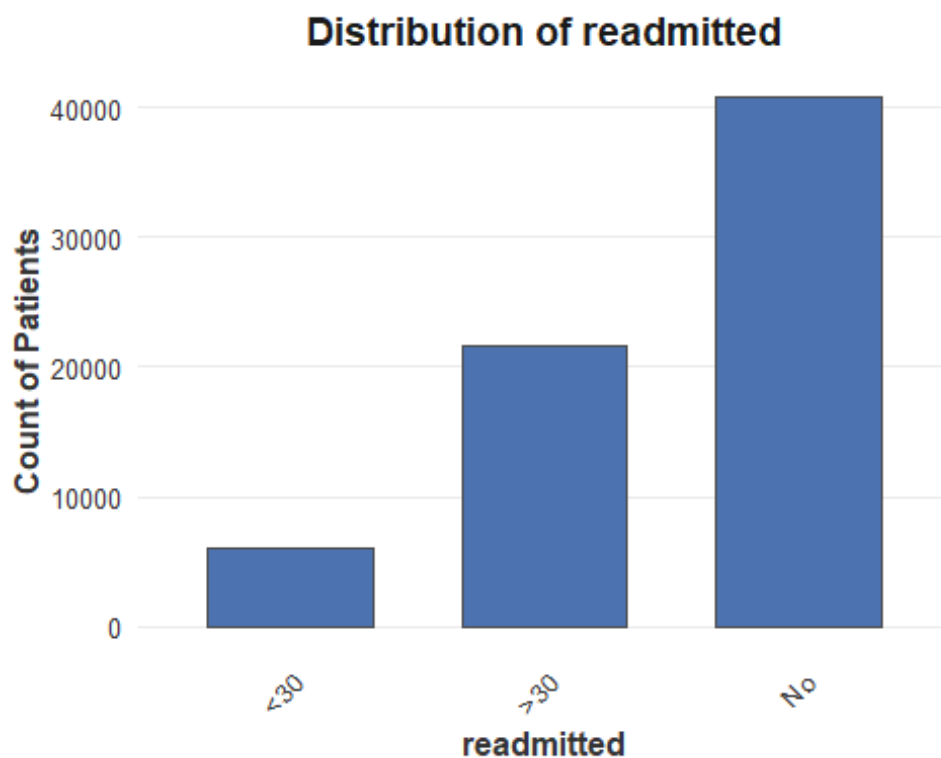


Fig.1. Distribution of readmissions by patient count

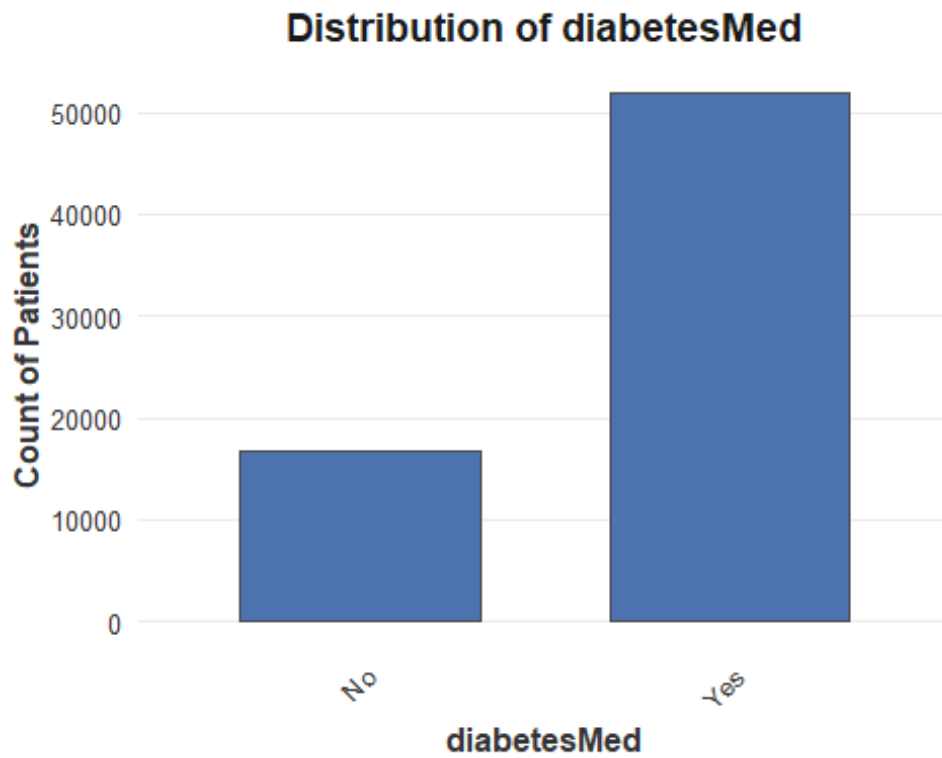


Fig.2. Distribution of Diabetes Medicine by patient count

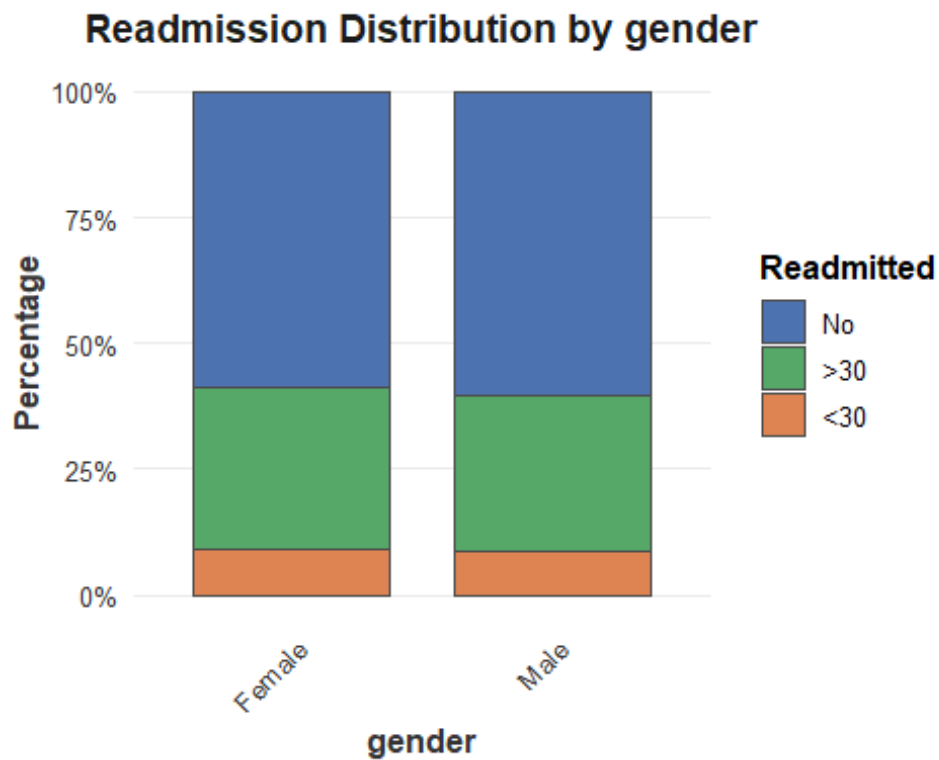


Fig. 3. Readmission distribution by gender

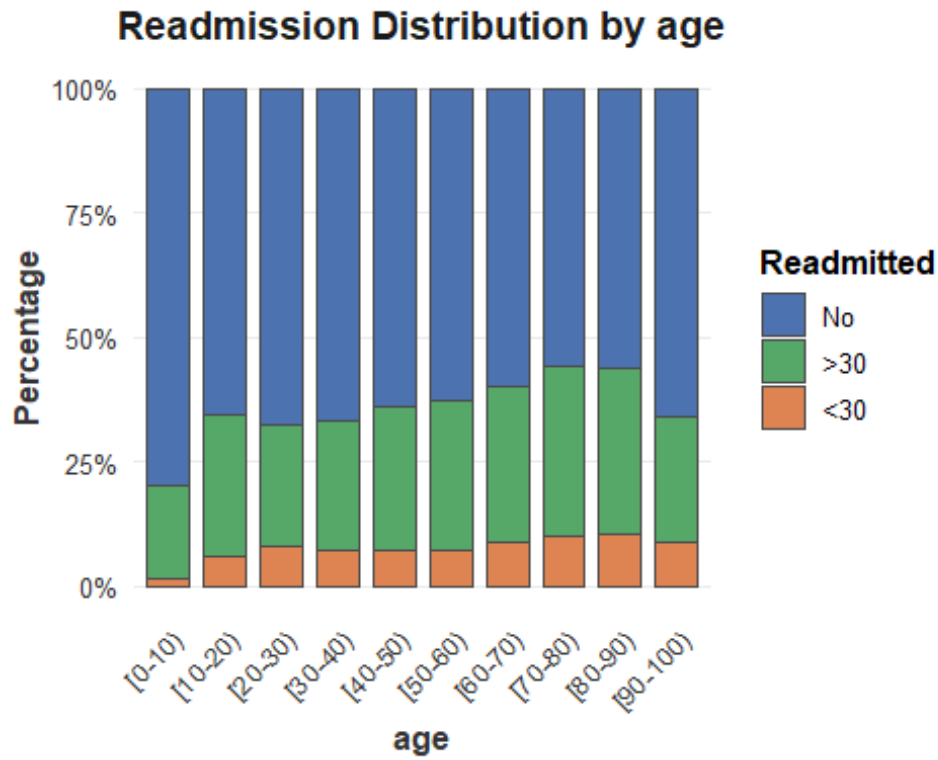


Fig. 4. Age distribution by readmission

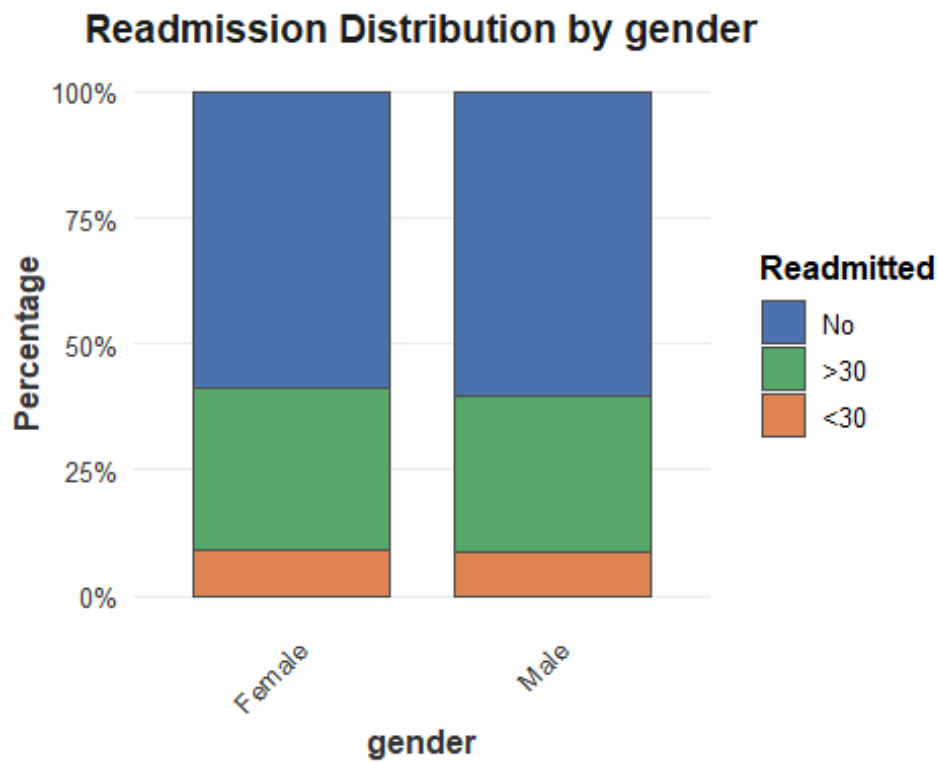


Fig.5. Distribution of readmissions by gender

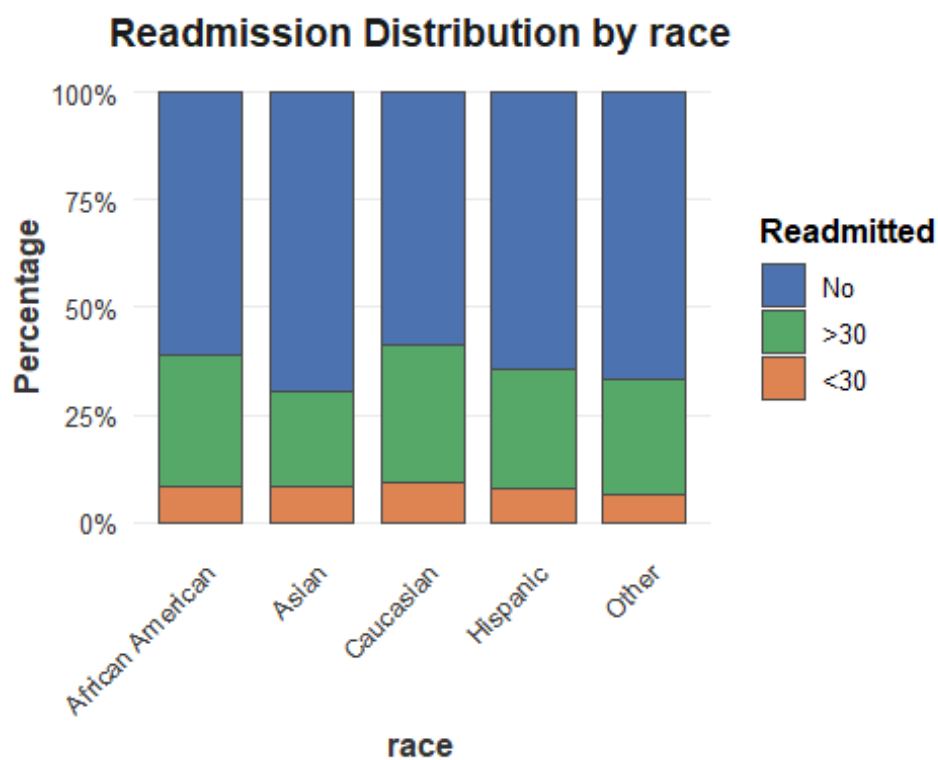


Fig.6. Distribution of readmissions by race

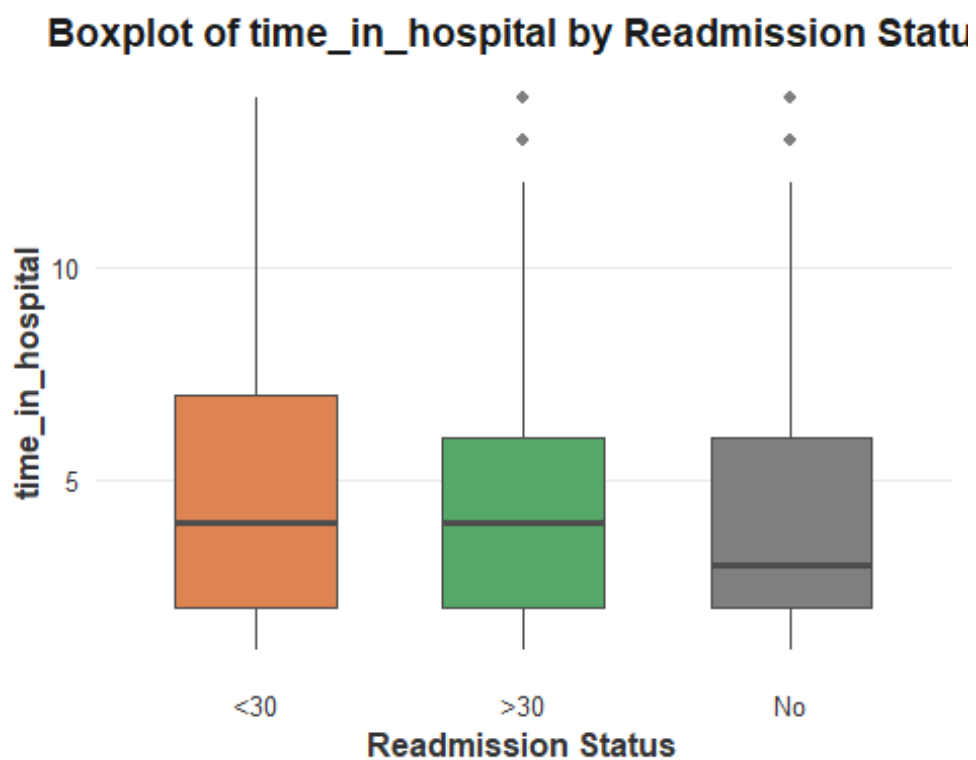


Fig.7. Boxplot of time in hospital by readmissions

Boxplot of num_medications by Readmission Status

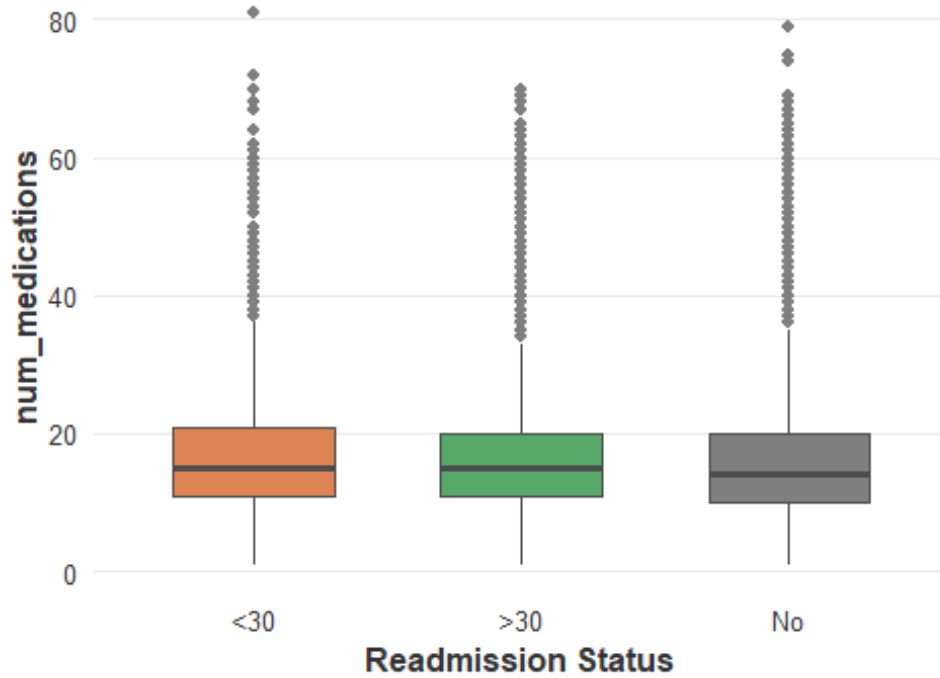


Fig.8. Boxplot of the number of medications by readmissions

Boxplot of num_lab_procedures by Readmission Status

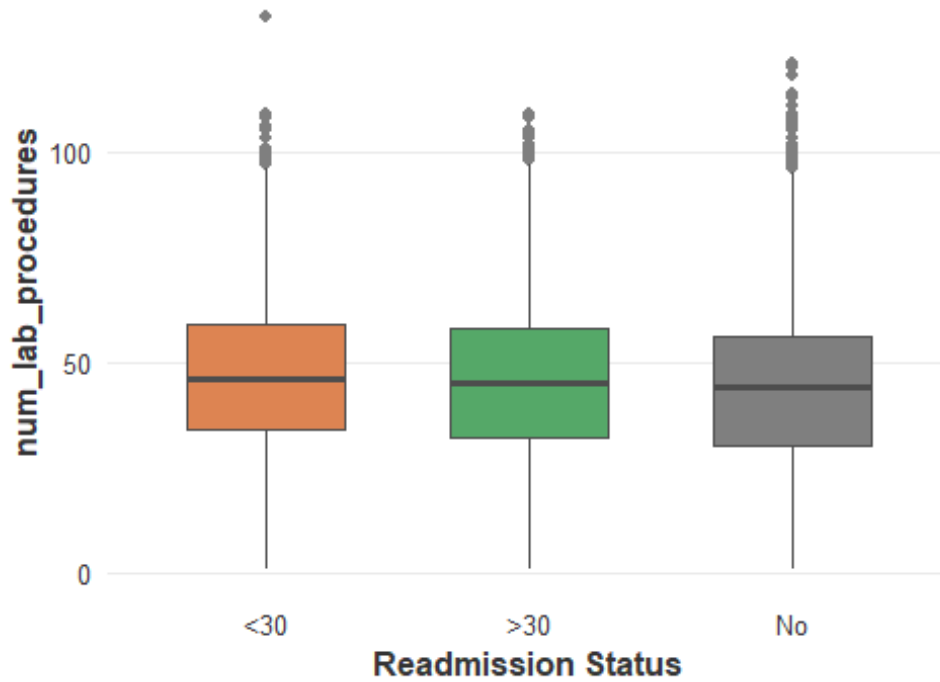


Fig.9. Boxplot of the number of lab procedures by readmissions

Scatterplot Matrix of Numerical Features

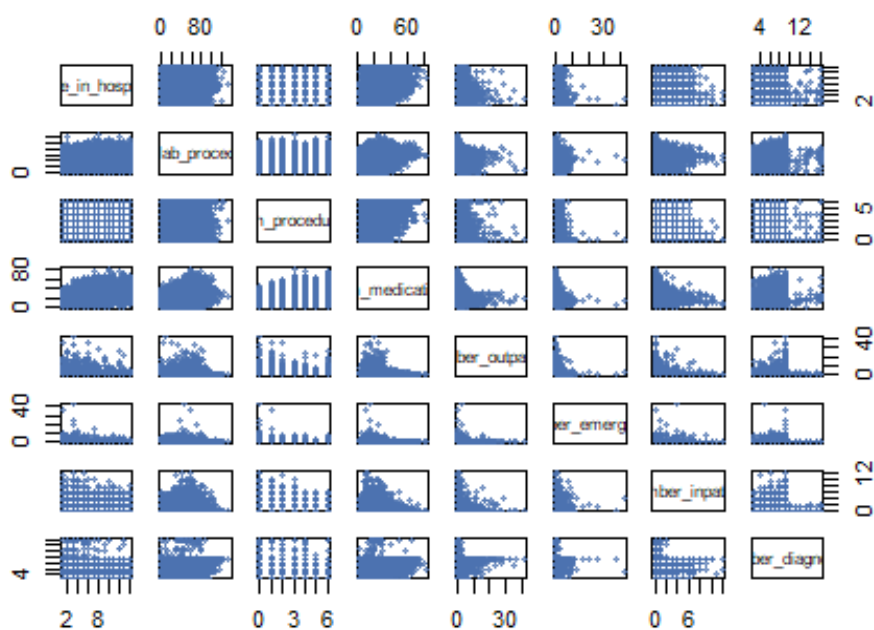


Fig.10. Scatterplot of numerical values

Correlation Heatmap of Numerical Features

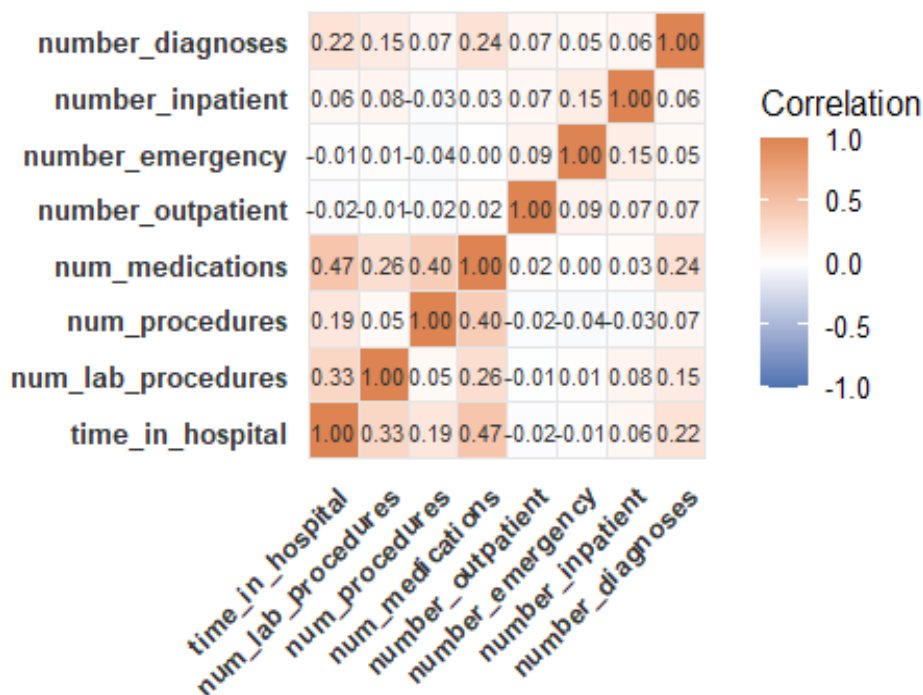


Fig.11. Correlation heatmap for numerical values

Table 1. Summary statistics of numerical values

<i>Variable</i>	<i>Mean</i>	<i>Median</i>	<i>Min</i>	<i>Max</i>
<i>Time in Hospital</i>	4.32	4	1	14
<i>Num. Lab Procedures</i>	43.13	44	1	132
<i>Num. Procedures</i>	1.44	1	0	6
<i>Num. Medications</i>	15.81	14	1	81
<i>Number Outpatient</i>	0.29	0	0	42
<i>Number Emergency</i>	0.11	0	0	42
<i>Number Inpatient</i>	0.18	0	0	12
<i>Number Diagnoses</i>	7.35	8	3	16

The distribution of readmissions reveals that there is a class imbalance between the subgroups of readmission within, beyond 30 days and no readmission. This pattern directly correlates with the diabetes medication distribution, which shows that the majority of patients are on diabetes medication. The readmission distribution by gender shows similar patterns between males and females, the age distribution shows a significant difference between age groups, and the race distribution shows only subtle variations between different races. The boxplot shows that the time in hospital that patients readmitted within 30 days tend to have longer initial hospital stays, while the medication boxplot shows that the same type of patients have more medications, and a higher number of lab procedures. The scatterplot and the correlation heatmap show that the numerical values have low to moderate correlations, which we do not need to worry about while developing our models, and the summary statistics table shows additional information for each one of the numerical values.

During the data visualisation and exploratory data analysis phase, some columns with almost zero variance were identified, which will be dropped later during our feature engineering and feature selection phase before starting our modelling. As these features offer no discriminative power for the predictive models, they were excluded from visual analysis.

Proposed Future Analysis

- Week 1 of 2nd Semester: perform the feature engineering to ensure data is ready for data reduction, clustering, and modelling.
- Week 2 to 3 of 2nd Semester: perform data reduction to reduce the data using different techniques, and then choose the best reduced data to perform cluster analysis on it.
- Week 3 to 4 of 2nd Semester: perform clustering on the best reduced data to uncover clusters and to add them as a feature to help in predictions.
- Week 4 to 8 of 2nd Semester: perform predictions using a two-stage hurdle model, the first stage will predict whether a patient is readmitted or not, and the second stage will predict early vs late readmission on conditional readmission occurring, this one will be done using various machine learning models.
- Week 8 to the end of the 2nd Semester: ensure the models are working properly and keep attempting to enhance their performance, along with writing the final thesis document.

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Appendix A:



DUNDALK INSTITUTE OF TECHNOLOGY School of Informatics & Creative Arts Ethical Approval Form for Research Projects

Researcher Name: Israa Atike

Year: 2025/2026

Title of project: End-to-End Analysis of Diabetes Patients: Data Cleaning, Data Visualisation, and Multi-Target Machine Learning Prediction

Name of supervisor/s: Damian Loughran
(if applicable)

Date: 24/10/2025

This application is to be completed by the researcher, and where appropriate, in conjunction with the project supervisor. The lead researcher/supervisor is responsible for submitting the completed form to the appropriate Research Ethics Committee (details below)

Please note: If your submission is incomplete or unclear, your application will be returned to you, and your project may be delayed.

Section 1

Type of Researcher

Please tick the appropriate box below to indicate the type of researcher you are:

Undergraduate
(proceed to section 2)

☐

Completed Ethical Approval forms for undergraduate research should be submitted to the relevant Departmental Research Ethics Committee (DREC).

Drec.dcam@dkit.ie – Department of Creative Arts, Media and Music
Drec.dcs@dkit.ie – Department of Computing Science and Mathematics
Drec.dvhcc@dkit.ie – Department of Visual and Human Centred Computing

Postgraduate
(proceed to section 3)

☐

Completed Ethical Approval forms for Postgraduate research should be submitted directly to the School Research Ethics Committee (SREC).

Srec.ica@dkit.ie

Staff
(proceed to section 3)

☐

Completed Ethical Approval forms for Staff research should be submitted directly to the School Research Ethics Committee (SREC).

Srec.ica@dkit.ie

Section 2

Please complete questions 1-4 listed below.

	Human and / or Animal Research	YES	NO
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1	Does your research involve human participants other than the following¹? • Research using exclusively secondary sources. • Research using materials legally accessible to the public that have legal protection, e.g., record of court judgements, data archives. • Research using materials that are publicly accessible and where there is no reasonable expectation for privacy, e.g., books, published third party interviews. • Observations of human behaviour in public where (i) those being observed have no reasonable expectation of privacy, (ii) there is no intervention on the part of the researcher nor any interaction between the researcher and those observed, and (iii) individuals are not identifiable in the results. If 'YES', please complete B and C below.		No
2	Does your project involve working with animals? – If 'YES', please complete B and C below. – Please note that for ethical consideration: 'Animals' are classed as vertebrate animals including cyclostomes and cephalopods (DIRECTIVE 2010/63/EU)		No
3	Does your project involve working with participants from any of the following categories? • Minors (under 18 years of age) • People with learning or communication difficulties • Patients • People in custody • People engaged in illegal activities If 'YES', please complete D below.		No
4	Does your project have any possible ethical implications other than those outlined in questions 1, 2 and 3? If 'YES', please complete E below.		No

A. I consider that this project has no significant ethical implications² to be brought through the ICA School Ethics Review Process *Please complete Section 4*

NA

¹ If there is any doubt, researchers should contact the Chair of the SREC.

² In determining significant implications, please consider all potential risks attached to this project. If there is any doubt, researchers should contact the Chair of the SREC.

B.

I. Is this study part of a larger project that already has ethical clearance?

YES / **NO**

If **YES** please answer question B II.

If **NO** please answer question C.

II. If this study is part of a larger project that already has ethical clearance, are you proposing any changes to the operational plan already ethically approved?

YES / NO **N/A**

If **YES**, please complete *Sections 3 and 4*.

If **NO**, then please provide the project details below and complete *Section 4*.

Title of project with ethical clearance: _____ **N/A** _____

C. Could this project have ethical implications that should be brought before the appropriate ICA Departmental Ethics Review Committee as it will be carried out with human participants?

YES / **NO**

If **YES**, please complete *Sections 3 and 4*.

If **NO**, please complete *Section 4*.

D. I consider that this project may have ethical implications that should be brought before the School Research Ethics Committee as it will be carried out with human participants in a “vulnerable” category.

Please complete Sections 3 and 4

<u>N/A</u>

All research carried out with human participants in a vulnerable category must be referred by the Departmental Research Ethics Committee to the School Research Ethics Committee for approval.

E. Could this project have ethical implications, other than those previously outlined, that should be brought before the appropriate ICA Departmental Ethics Review Committee?

YES / NO N/A

If **YES**, please complete *Sections 3 and 4*.
If **NO**, please complete *Section 4*.

Section 3

3.1 Application Form Checklist

Please complete Section 3 and provide additional information as attachments.

My application includes the following documentation:	INCLUDED (mark as YES)	NOT APPLICABLE (mark as N/A)
Recruitment advertisement		NA
Participant Information Leaflet		NA
Participant Informed Consent form		NA
Questionnaire/Survey		NA
Interview/Focus Group Questions		NA
Debriefing material		NA
Evidence of approval to gain access to off-site location		NA
Ethical approval from external organizations. If ethical approval from external organizations is pending give details below		NA

Details
NA

3.2 Project Details

a) Lay description (Maximum 200 words)

Please outline, in terms that any non-expert would understand, what your research project is about, including what participants will be required to do. Please explain any technical terms or discipline-specific phrases.

My study titled “End-to-End Analysis of Diabetes Patients: Data Cleaning, Data Visualization, and Multi-Target Machine Learning Prediction”, will include most importantly the data cleaning to handle missing values, inconsistent entries, and irrelevant variables, as well as exploratory data analysis and data visualization to reveal patterns, and correlations and finally multi-target machine learning models, which will predict more than one variable, my study will differ from other studies in the multi-target predictions and comprehensive analysis as well as the use of new Machine Learning models, rather than using a single-target prediction and outdated statistical models. To enhance the practical impact of this project, an interactive Shiny web application will be developed to present the findings and predictions in an accessible and user-friendly manner, will allow users such as clinicians, hospital administrators, and researchers to interact with the dataset, explore demographic and clinical trends, and visualize patterns related to diabetes management, users will also be able to input patient details to obtain real-time predictions for outcomes such as readmission likelihood, medication changes, and lab result anomalies. **Note that the project will not gather any sensitive personal information about the participants.**

b) Research objectives (Maximum 150 words)

Please summarise the objectives of the research.

The main objective of this research is to perform an end-to-end analysis of diabetes patient data to improve understanding and prediction of patient outcomes. Specifically, the project aims to clean and preprocess the dataset to ensure accuracy and reliability, perform exploratory data analysis and visualization to uncover meaningful trends, and develop multi-target machine learning models

capable of simultaneously predicting multiple outcomes, such as 30-day readmission, medication changes, and lab result anomalies. The study also aims to identify key demographic, clinical, and treatment-related factors influencing these outcomes. By comparing the performance of different machine learning models, the research seeks to determine the most effective predictive approaches. Ultimately, the project strives to contribute to data-driven healthcare insights that can support clinicians in decision-making and improve patient care for individuals with diabetes.

c) Research location and duration

Location(s)/Population*	
Research start date	20 October 2025
Research end date	10 May 2026
Approximate duration	6 months and half

* If location/Population other than DkIT campus/population, provide details of the approval to gain access to that location/population as an appendix.

3.3 Participants

		YES	NO	N/A
Do participants fall into any of the following special groups?	Minors (under 18 years of age)			x
	People with learning or communication difficulties			x
	Patients			x
	People in custody			x
	People engaged in illegal activities (e.g. drug-taking)			x
Have you given due consideration to the need for satisfactory Garda clearance?				x

3.4 Sample Details

Approximate number	NA
Where will participants be recruited from?	NA
Inclusion Criteria	NA
Exclusion Criteria	NA
Will participants be remunerated, and if so in what form?	
NA	

Justification for proposed sample size and for selecting a specific gender, age, or any other group if this is done in your research.
NA

3.5 Risk to Participants

- a) Please describe any risks to participants that may arise due to the research. Such risks could include physical stress, emotional distress, perceived coercion e.g. lecturer interviewing own students. Detail the measures and considerations you have put in place to minimize these risks
- b) What will you communicate to participants about any identified risks? Will any information be withheld from them about the research purpose or procedure? If so, please justify this decision.

3.6 Informed Consent

	YES	NO	N/A
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Will you obtain active consent for participation?			X
Will you describe the main experimental procedures to participants in advance?			X
Will you inform the participants that their participation is voluntary and may be withdrawn at any point?			X
If the research is observational, will you ask for their consent to being observed?			X
With questionnaires, will you give participants the option of omitting questions they do not want to answer?			X
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?			X
Will the data be anonymous?			X
Will you debrief participants at the end of their participation?			X
Will your project involve deliberately misleading participants in any way, or will information be withheld? If you answer yes, give details and justification for doing this below.			X
NA			

a) Please outline your approach to ensuring the confidentiality of data (that is, that the data will only be accessible to agreed upon parties and the safeguarding mechanisms you will put in place to achieve this.) You should include details on how and where the data will be stored, and who will have access to it.

NA

b) Please outline how long the data will be retained for, if it will be destroyed and how it will be destroyed.

Storage

Data will be stored in a password protected file. DkIT's OneDrive system will be used as a storage mechanism. Any hardcopy data needs secure storage onsite at DkIT.

Access

Only DkIT staff and the student will have access to data (student, supervisor etc.)

Communication

DkIT email accounts will be used for communications regarding data.

Digital Platform Usage

The only online digital platform that will be used is OneDrive.
The questionnaire will ask participants to provide consent to their data being housed by the platform.

Section 4

Researcher I have read and I understand the DkIT Ethics Policy available from:
<https://www.dkit.ie/assets/uploads/documents/Research/Policies/DkIT%20Research%20Ethics%20Policy.pdf>

Signed: IA
(Researcher)

Print Name: Israa Atike

Date: 16/10/2025

Supervisor: Applications for Ethical Approval of Undergraduate projects are forwarded to the Departmental Research Ethics Committee for approval or referral to the School Research Ethics Committee. Applications for Ethical Approval of Postgraduate and Staff projects are sent to the School Research Ethics Committee for Approval.

I have read and approved this form & information:

Signed: DL Print Name: Damian Loughran Date: 24/10/25
(Supervisor/Head of Department/ Research Centre Director/ Head of School)

There is an obligation on the researcher and/or supervisor to bring to the attention of the Departmental/School Research Ethics Committee(s): (a) Any issues with ethical implications not clearly covered by this form (b) Any ethical issues which may arise during the carrying out of the research; (c) Any ethically significant change made to the project after approval.

Section 5 (For office use only)

STATEMENT OF ETHICAL APPROVAL (FOR UNDERGRADUATE PROJECTS ONLY)

This project has been considered using agreed department procedures and is now:

Approved:

☐

Referred to the School Ethics Committee:

☐

Signed: _____ Print Name: _____ Date: _____

(Chair of Departmental Research Ethics Committee/Head of Department)

STATEMENT OF ETHICAL APPROVAL

This project has been considered using agreed School procedures and is now:

Approved:

☐

Rejected (further information sought):

☐

Chair of School Research Ethics Committee

This project has been considered by the Ethics Committee and ethical approval is granted.

Signed: _____

Print Name: _____ Date: _____

Chair of School Research Ethics Committee